

- How about the case with more than 2 periods?

For illustration, suppose we have n individuals and 3 periods.

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it} = \beta_0 + \beta_1 x_{it} + a_i + v_{it}, \quad i = 1, \dots, n; t = 1, 2, 3$$

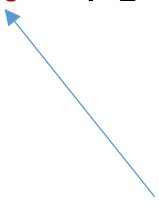
- The first differenced equation is

$$(t=2) - (t=1): y_{i2} - y_{i1} = \beta_1(x_{i2} - x_{i1}) + (v_{i2} - v_{i1})$$

$$(t=3) - (t=2): y_{i3} - y_{i2} = \beta_1(x_{i3} - x_{i2}) + (v_{i3} - v_{i2})$$

$$\Delta y_{it} = \alpha_0 + \beta_1 \Delta x_{it} + \Delta v_{it} \quad i = 1, \dots, n; t = 2, 3$$

In case $E(\Delta v_{it}) \neq 0$



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$$\Delta y_{it} = \alpha_0 + \alpha_1 \text{time_dummy} + \beta_1 \Delta x_{it} + \Delta v_{it} \quad i = 1, \dots, n; t = 2, 3$$

Use 1 dummy to differentia two groups (t=2 and 3)

